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Belt buckle system with threaded retainers

Abstract

Clothing belt adjustment systems are configured for wearing around a user's body such as around the waist. The belt adjustment system includes a belt and a buckle wherein the belt can be looped and secured to the buckle to secure the belt around a user's waist or other portion of the body. The buckle has a body with a thin side profile that is not overly bulky such that it can comfortably fit over the button region of a wearer's pants. The buckle advantageously includes one or more threaded retainer members, such as threaded screws, that secure one portion of the buckle to another portion of the buckle.

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Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
4823443	12/1988	Waters	N/A	N/A
5031956	12/1990	Hudgins	N/A	N/A
5806145	12/1997	Chen	N/A	N/A
5875523	12/1998	Chen	N/A	N/A
6137675	12/1999	Perkins	N/A	N/A
6145169	12/1999	Terzuola et al.	N/A	N/A
6295700	12/2000	Pizak	N/A	N/A
6427291	12/2001	Kim	N/A	N/A
6560825	12/2002	Maciejczyk	N/A	N/A
6698070	12/2003	Chen	N/A	N/A
6748630	12/2003	Livingston	N/A	N/A
8590114	12/2012	Shindo	N/A	N/A
8689364	12/2013	Rowland et al.	N/A	N/A
9145932	12/2014	Xiao	N/A	N/A
9149090	12/2014	Taylor	N/A	N/A
9210971	12/2014	Catanese et al.	N/A	N/A
9277775	12/2015	Lopez	N/A	N/A
9345288	12/2015	Greenberg et al.	N/A	N/A
9560898	12/2016	Hortnagl	N/A	N/A
9961967	12/2017	Costenbader	N/A	N/A
10208778	12/2018	Hunsucker	N/A	A41F 9/002
10321752	12/2018	Zentil et al.	N/A	N/A
10433619	12/2018	Wong et al.	N/A	N/A
10548362	12/2019	Hunsucker et al.	N/A	N/A
10681963	12/2019	Rowland et al.	N/A	N/A
10718406	12/2019	Farrell	N/A	N/A
10881170	12/2020	Hunsucker et al.	N/A	N/A
10926114	12/2020	Thompson et al.	N/A	N/A
10966489	12/2020	Rowland et al.	N/A	N/A
10974638	12/2020	Fountain, III	N/A	N/A
11019887	12/2020	Huang	N/A	N/A
11083250	12/2020	Mowad	N/A	N/A
2010/0011487	12/2009	Kim	N/A	N/A
2013/0305493	12/2012	Hortnagl	N/A	N/A
2019/0053581	12/2018	Hestvik et al.	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
2382206	12/1977	FR	N/A

Background/Summary

REFERENCE TO CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a divisional of U.S. patent application Ser. No. 15/999,564, filed Aug. 20, 2018, which claims priority to and benefit of U.S. Provisional Application No. 62/547,835, filed Aug. 20, 2017, the entire disclosures of which are hereby incorporated by reference herein.

BACKGROUND

(1) Conventional belt buckles or belt adjustment systems are limited in their ability to conform to a particular user's waist size. Belt adjustment systems conventionally secure a belt about a user's waist by relying on a series of spaced holes punched through an end of a belt. A hook of a belt buckle can be inserted through a hole to capture the end of the belt to secure the belt in a loop of a particular size. The spacing between each of the holes as well as the overall number of holes can vary for adjustment of belt size, but is generally limited by the minimal material that must remain between the holes.

(2) Conventional belt adjustment systems are limited to setting the size of the belt loop to discrete sizes based upon the spacing of the holes in the belt. If a user desires to set the belt to a loop size that is positioned between the holes in the belt, the user has to manually create an additional hole in the belt, which can be difficult and unattractive if not performed well. Alternately, the user must use the next smaller or next larger belt loop size relative to the desired size, which can be uncomfortable for the user.

(3) Other belt adjustment systems are known that increase the flexibility of adjustment to a variety of waist sizes, but these adjustment systems are not typically fashionable or aesthetically pleasing.

SUMMARY

(4) There is a need for a belt adjustment system that permits a continuum of belt loop sizes or a larger selection of belt loop sizes. Disclosed herein are clothing accessories, particularly belt adjustment systems for wearing around a user's body such as around the waist. The belt adjustment system includes a belt and a buckle wherein the belt can be looped and secured to the buckle to secure the belt around a user's waist or other portion of the body.

(5) In an embodiment, the buckle includes a front face that is movable between an open and closed position to reveal or hide, respectively, a chamber within the buckle. The chamber is sized and shaped to contain one or more items when the front face is closed. In an embodiment, the items include at least one golf ball marker and at least one golf divot repair tool that also serves as a bottle opener. The buckle has a body with a thin side profile that is not overly bulky such that it can comfortably fit over the button region of a wearer's pants. The buckle advantageously includes one or more threaded retainer members, such as threaded screws, that secure one portion of the buckle to another portion of the buckle.

(6) Other features and advantages should be apparent from the following description of various embodiments, which illustrate, by way of example, the principles of the claimed subject matter.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) These and other aspects will now be described in detail with reference to the following

drawings.

(2) FIG. 1 shows a schematic view of an embodiment of a belt system;

(3) FIG. 2 shows a perspective, side view of a belt buckle of the belt system.

(4) FIG. 3 shows a bottom view of the belt buckle.

(5) FIG. 4 shows a bottom view of the belt buckle.

(6) FIG. 5 shows the belt buckle with a door in an open state such that an internal chamber is exposed.

(7) FIG. 6 shows the belt buckle with a door in an open state such that an internal chamber is exposed and components removed from the internal chamber.

(8) FIG. 7 shows an example of a belt container in which the belt can be stored.

(9) FIG. 8 shows the belt container in an open state.

(10) FIG. 9 shows a side view of another embodiment of a buckle with threaded retainer members.

DETAILED DESCRIPTION

(11) Before the present subject matter is further described, it is to be understood that this subject matter described herein is not limited to particular embodiments described, as such may of course vary. It is also to be understood that the terminology used herein is for the purpose of describing particular embodiments only, and is not intended to be limiting. Unless defined otherwise, all technical terms used herein have the same meaning as commonly understood by one skilled in the art to which this subject matter belongs.

(12) FIG. 1 shows a perspective view of an embodiment of a belt system **100** that may be worn with a pair of pants, shorts, trousers, skirts or other articles of clothing. The system **100** can also be used with other items such as watch straps, purse straps, guitar straps or animal collars or other articles that may include a buckle system that is adjusted for size or where a number of size variations would be desirable. Some figures include exemplary numerical dimensions. It should be appreciated that the dimensions are for example only and are not intended to be limiting. The belt buckle system can be configured with dimensions outside of the ranges and values shown.

(13) The belt system **100** includes an elongated belt **105** and a buckle **110** disposed on a first end of the belt **105**. The buckle **110** is removably attached to the belt such as by using a clamp on the buckle **110**. The belt buckle **110** is a pinless buckle in that it does not use a pin to secure itself to the belt **105** when the belt is looped around a user's waist. The belt buckle **105** has a front face that faces away from a user or a user's torso when the belt is worn around the user's waist. That is, the belt buckle **105** sits flat against the user's waist or torso such that the buckle will be positioned over the region of the user's pants where a button is typically located on the user's pants.

(14) In an example embodiment, the belt **105** is coupled to a holster that is sized and shaped to hold a firearm, such as a pistol or gun.

(15) FIG. 2 shows a perspective, side view of the belt buckle **110**. FIGS. 3 and 4 show bottom views of the belt buckle **110**. The belt buckle **110** has a front face that is formed by a door **205** or other movable portion. The door **205** is movable relative to a body **207** of the belt buckle such that the door **205** can be opened (such as in a pivoting manner) relative to the body to expose an internal chamber, as described in more detail below. A belt loop structure **225** is attached to a main body of the buckle **110** such as at one or more attachment locations **230**. In an embodiment, the belt loop structure **225** can pivot or rotate about an axis that intersects the attachment location **230**. In addition, as described more fully below, at least one threaded member, such as a threaded screw, extends through the belt loop structure **225** and the main body of the buckle **110** to secure the belt loop structure **225** to the main body.

(16) With reference to FIGS. 3 and 4, the buckle **110** has a bridge **350** that forms or at least partially defines an opening through which the opposite end of the belt can be threaded when worn by a user. A movable latch **310** or other pinless member is configured to move towards and engage with a portion of the opposite end region of the belt when the opposite region is positioned through the opening formed by the bridge **350** for securing the belt to the buckle in a desired position. In an

embodiment, the latch **310** secures to a tooth or other engagement portion formed within a row of engagement portions on the belt. The latch **310** can be biased toward a latching engagement with the belt such as by using a biasing member, which can be, for example, a magnet, a spring, or other device.

(17) As mentioned above with reference to FIG. 2, at least one threaded member, such as a threaded screw, extends through the belt loop structure **225** and the main body of the buckle **110** to secure the belt loop structure **225** to the main body. A threaded member such as a threaded screw can also be used to secure other portions of the buckle **110** to one another. For example, a threaded screw can be used to secure the bridge **350** to the main body of the buckle. This is different and more secure than using a non-threaded pin to secure the buckle portions to one another. A non-threaded pin has a tendency to loosen from the buckle such that the buckle can become unstable or fall apart at some point during use.

(18) FIG. 9 shows a side view of the belt buckle. One or more threaded members, such as threaded screws **905**, secure portions of the belt buckle to one another. For example, threaded screws **905** secure the bridge **350** and/or the belt loop structure **225** to the main body of the belt buckle. In this regard, one or more aligned, threaded openings or passageways can extend through the bridge **350** and the main body. A screw **905** can be threadedly positioned in the aligned passageways to secure the components of the buckle to one another via the screw **905**. It should be appreciated that threaded screws can be used to secure other portions of the buckle to one another. In addition, the threaded screw can have various types of heads, such as Allen heads, for attaching to a drive member, such as an Allen wrench.

(19) The threaded screw **905** is an elongated body such as a cylindrical body having a head on a proximal end and threads on an external surface of a distal region of the body or over the entire body. The head portion of the screw may be enlarged in diameter relative to a remainder portion of the screw or may have the same diameter as the remainder portion of the screw. The screw **905** is an externally threaded fastener capable of being inserted into holes in any portion of the buckle for securing a first component of the buckle to any other component or components of the buckle. The screw is capable of mating with a preformed internal thread or forming its own thread within the respective hole, and of being tightened or released by torquing the head. In an example embodiment, the screw has a diameter in the head and/or body portion of 3 mm. In another embodiment, the screw has a diameter of 1-2 mm or 1-3 mm.

(20) It should be appreciated that threaded screws are much more secure than pins or other types of non-threaded structures. The threaded screws can withstand forces that the buckle undergoes during use. The buckle can experience high levels of torque and linear forces, which can cause the buckle to come apart or otherwise malfunction. The threaded screws **905** eliminate or greatly reduce the likelihood that the belt buckle will come apart as a result of such forces.

(21) In a method of manufacture, one or more of the components of the buckle are provided. The components are assembled in a manner that forms the shape of the buckle. One or more of the components can be drilled or tapped with appropriate sized holes, openings, or passageways that are sized and shaped to receive a corresponding screw. One or more of the holes, openings, or passageways can be threaded. The appropriate sized screw can then be inserted into a corresponding opening and threadedly retained in the opening to secure one or more components of the buckle to other components of the buckle.

(22) With reference now to the side view of FIG. 2, the body **207** of the buckle **110** is shaped such that a cavity or cutout **215** is formed on the side of the body **207**. The cutout **215** is such that a region of the body **207** is thinner relative to an adjacent region, with the region being thinner along a dimension or direction normal to a wearer's body when the belt is worn around the wearer's waist. That is, the direction is a direction along the line of sight of a person that is facing the wearer and looking toward the front side of the wearer. In this manner, the body of the buckle is so dimensioned so that the wearer's pants button does not contribute to or cause the belt buckle to

protrude any further outward from the wearer's pants if the wearer's pants did not have a button in the region of the buckle when the buckle is worn. The cutout is so dimensioned relative to a wearer's pant button so that the pant button fits within the cutout.

(23) The cutout **215** is sized and shaped to form a gap in the body of the buckle in which the wearer's pants button can be positioned when the belt is worn. In this manner, the buckle can be positioned atop the button without the button contributing to the overall size of the buckle or pushing the buckle outward away from the user's body. The button therefore does not interfere with the belt buckle and does not result in the buckle being pushed outward from the user's body when the belt is worn around the waist. FIG. 2 shows one side view of the belt buckle. It should be appreciated that the opposite side view of the belt buckle also has a cutout **215** similar to the cutout **215** shown in FIG. 2. The cutout **215** is located along the length of the belt buckle with the length being the longitudinal direction of the belt member. The cutout **215** extends along only a portion of the entire length of the belt buckle such that the belt buckle has a thickness that is greater where the cavity or cutout **215** is not located relative to whether cavity or cutout **215** is located.

(24) FIG. 5 shows the belt buckle **110** with the door **205** in an open state such that the internal chamber is exposed. The internal chamber is exposed and accessible when the door **205** is open as shown in FIG. 5. When the door **205** is closed (as in FIGS. 1 and 2 for example), the door **205** is positioned relative to the belt buckle body such that the internal chamber is covered and hidden by the door **205**. In this manner, the internal chamber is collectively formed by the body of the belt buckle and the door **205** when the door is closed.

(25) The door **205** may move relative to the body of the belt buckle in a variety of manners including in a pivoting or rotating manner. In this regard, the one or more hinges attach the door **205** to the belt buckle body. Other types of movement are possible such as a sliding movement between the door **205** and the body.

(26) As shown in FIG. 5, the internal chamber is sized and shaped to contain one or more items. The items are covered or hidden when the door **205** is closed. In the illustrated embodiment, the items include a ball marker **505**, which is disc-like member. The ball marker is positioned within a complementary-shaped indentation on the inner side of the door **205** and can be secured thereto such as by a magnet for example. The items also include a golf divot repair tool **510** that sits on the body **207** in the internal cavity. The divot repair tool **510** is secured within the cavity by a crossing member **520** that fits over the divot repair tool and secures it therein such as in a press fit manner. Other retaining elements can also be used such as one or more magnets to secure the divot repair tool **510** or other items in the chamber. FIG. 6 shows the buckle **110** with the ball marker **505** and divot repair tool **510** removed from the internal chamber.

(27) With reference to FIG. 5, the door **205** can be secure in the closed position such as by using one or more magnetic engagements **507** between the body **207** of the buckle and the door **205**. In this regard, a magnet may secure in the door **205** and/or the body **207** so that they contact one another and secure the door in the closed position when the door is closed. The magnet(s) may be positioned and secured beneath a securing element, such as a bridge structure **350**. Other ways of securing the door in the closed position can be used.

(28) In an embodiment, the belt **100** can be stored in a belt container **710**, as shown in FIGS. 7 and 8. The belt container **710** is sized and shaped to define an interior cavity that can contain the belt **100** in a rolled state. The interior cavity of the container **710** is sized so that the belt **100** is snugly stored within the container **710** such that the belt will not rattle or move when the container is closed. In this regard, the container **710** has a door that can be opened (as shown in FIG. 8) and closed (as shown in FIG. 7) to provide access to the belt.

(29) While this specification contains many specifics, these should not be construed as limitations on the scope of an invention that is claimed or of what may be claimed, but rather as descriptions of features specific to particular embodiments. Certain features that are described in this specification in the context of separate embodiments can also be implemented in combination in a single

embodiment. Conversely, various features that are described in the context of a single embodiment can also be implemented in multiple embodiments separately or in any suitable sub-combination. Moreover, although features may be described above as acting in certain combinations and even initially claimed as such, one or more features from a claimed combination can in some cases be excised from the combination, and the claimed combination may be directed to a sub-combination or a variation of a sub-combination. Similarly, while operations are depicted in the drawings in a particular order, this should not be understood as requiring that such operations be performed in the particular order shown or in sequential order, or that all illustrated operations be performed, to achieve desirable results. Only a few examples and implementations are disclosed. Variations, modifications and enhancements to the described examples and implementations and other implementations may be made based on what is disclosed.

(30) Although embodiments of various methods and devices are described herein in detail with reference to certain versions, it should be appreciated that other versions, embodiments, methods of use, and combinations thereof are also possible. Therefore, the spirit and scope of the appended claims should not be limited to the description of the embodiments contained herein.

Claims

1. A clothing belt system, comprising: an elongate belt member comprising a first end, a second end and an inner surface near the second end, wherein the inner surface has a series of teeth; a buckle on a first end of the belt member, the buckle adapted to engage the series of teeth of the belt member in a pinless manner such that the belt does not require holes and the buckle does not require a pin, the buckle being formed of at least two components; at least one threaded screw that retains a first component of the buckle to a second component of the buckle.
 2. The belt system as in claim 1, wherein the at least one threaded screw has a diameter of 1 mm.
 3. The belt system as in claim 1, wherein the buckle includes a main body and a bridge that forms an opening through which the second end of the belt can be threaded when worn by a user.
 4. The belt system of claim 1, wherein the threaded screw retains the main body to the bridge.
 5. The belt system as in claim 4, further comprising a door on the belt buckle that opens and closes to reveal a hidden compartment.
 6. The belt system as in claim 5, wherein the compartment is sized and shaped to contain at least one of a ball marker and a divot repair tool.
 7. The belt system as in claim 5, wherein the door to separate from the body of the belt buckle.
 8. The belt system as in claim 5, wherein the door is pivotably attached to the body of the belt buckle.
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